EEE 228 Wearable, optical, heart rate monitor individual report

1.Introduction

1.1 Context

The increasing prevalence of smart technology has led to the widespread use of wearable devices. Apple recently released the APPLE WATCH S9, which not only enhances the accuracy of daily health tests such as heart rate and oxygen saturation but also incorporates various daily tools. Monitoring your heart rate can provide valuable health information such as baseline amounts of exercise, mood changes, heart health levels and more. Reaching 81 million units by 2021, indicating a 16% sales share of the total wearable device market. Poongodi, T., Krishnamurthi, R., Indrakumari, R., Suresh, P., Balusamy, B. (2020)..

1.2 Specific aim for exercise

This exercise involves exploring the technology behind a health monitoring system and designing and assembling a sensor that is as accurate as possible in real-world scenarios.

1.3 Criteria for success

Success is evaluated by comparing it to existing high-precision sensors, with a smaller percentage error indicating closer success.

2. Theory

2.1 Photoplethysmography principles

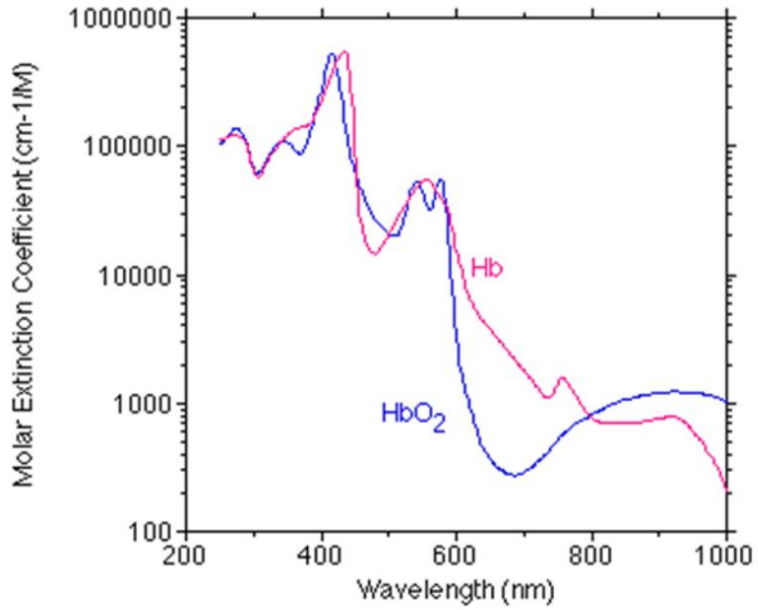


Figure 1 - specific extinction coefficient spectra of oxy and deoxyhaemoglobin. Note the logarithmic y axis.

$\boldsymbol{log_{10}\left(\frac{I_0}{I}\right) = \varepsilon * C * d}$ here I_0 represents the incident light intensity. I is the detected light intensity.d is the distance through the sample. C is the concentration and ε is the specific extinction coefficient. The figure above shows that blood samples with and without oxygen have different coefficients for the same wavelength of light. Only need to focus on one particular state for this. This leads to the second equation.

$\boldsymbol{log_{10}\left(\frac{I_{Max}}{I_{Min}}\right) = \varepsilon * \Delta C * d}$ In essence, a heartbeat can be measured by taking the logarithm of the light intensity measurement after dividing it back and forth while keeping the

measurement distance and coefficient unchanged. This implies that the frequency of the change in current due to different changes in light intensity is measured, which in turn calculates the frequency of the heartbeat.

2.2 LED: physical and circuit principles

Light-emitting diodes (LEDs) are devices that emit light when an electric current is passed through the PN junction of a diode. This process produces both visible and non-visible light. The circuit design for the LED is relatively straightforward. It requires the addition of a resistor in series with the Arduino's 5V supply, in forward bias mode. The resistor serves three main purposes: to limit the current, protect the circuit, and maintain the LED in a proper operating state. $R = \frac{V_{Arduino} - V_{LED}}{I_{LED}}$.

2.3Photodetector: physical and circuit principles.

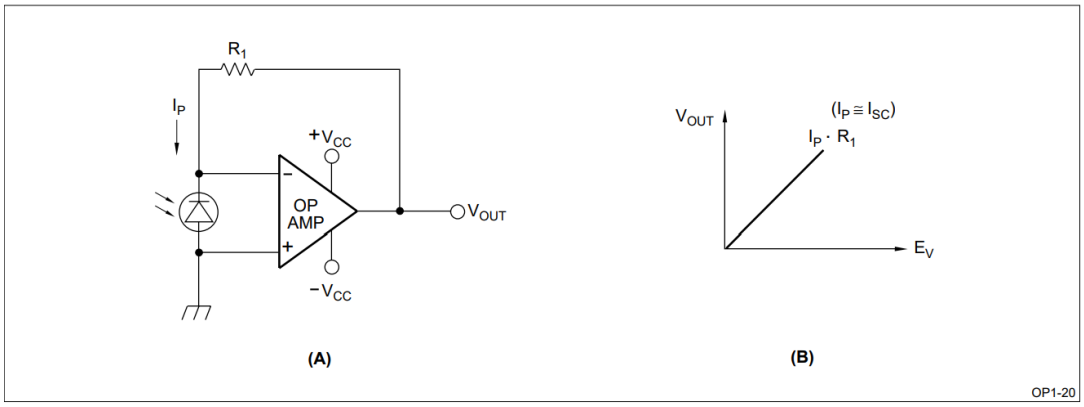
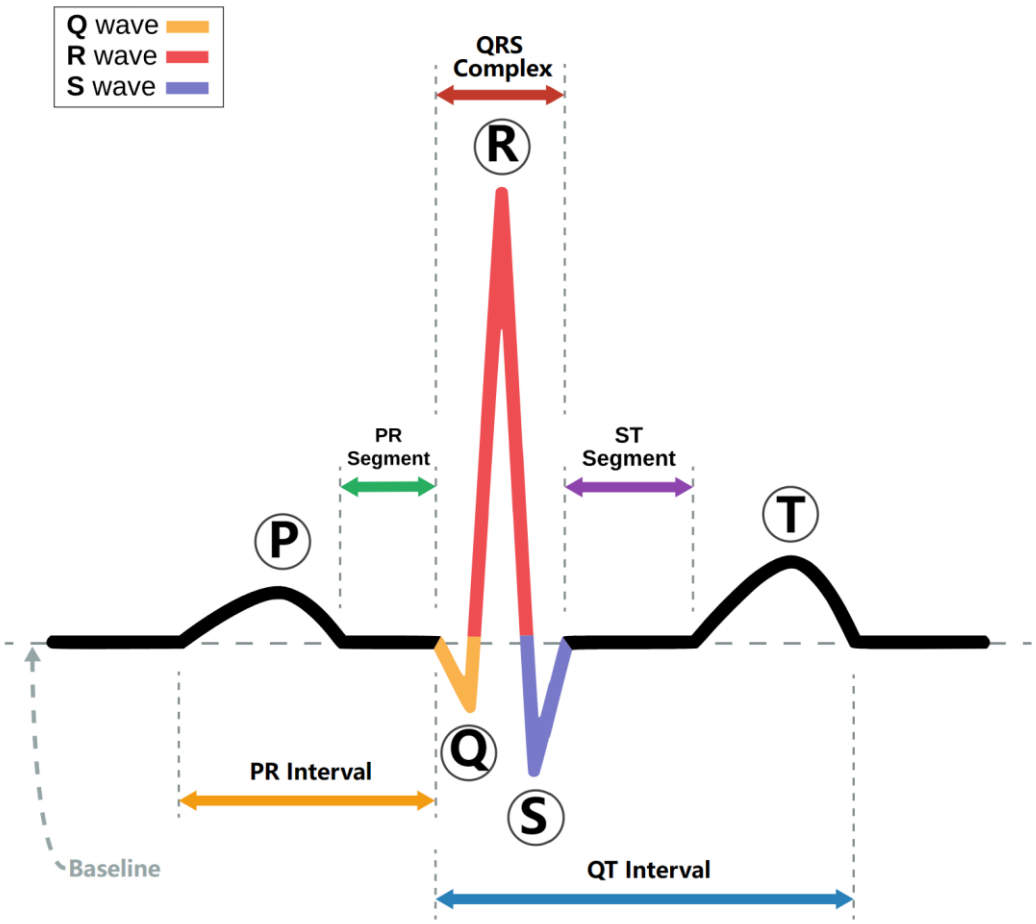


Figure 6. Photocurrent Amplifier using an Operational Amplifier (Without Bias)

A photodiode is a relatively conventional semiconductor component that converts optical signals into electrical signals. The principle of operation is based on the absorption of energy from external light, which generates carriers in the PN junction region. These carriers produce an electric current due to the built-in electric field. The circuit depicted in the accompanying figure illustrates the amplification of the photocurrent by the amplifier, which is subsequently converted to an optical voltage and presented on an oscilloscope or data acquisition (DAQ) system. It is noteworthy that, due to the amplifier's proximity to an ideal amplifier, the conversion of the photovoltage is governed by the product of current and resistance ($I * R$). Furthermore, the photodiode is biased in reverse, which enhances the generation of photocurrent.

2.4 Pulse detection algorithms



Wikipedia contributors, "Electrocardiography," *Wikipedia*, Apr. 09, 2024. https://en.wikipedia.org/wiki/Electrocardiography

The ECG heart rate graph above illustrates the various bands (P-wave, R-wave, etc.) that can be associated with the use of thresholding to calculate the heart rate. The principle is that, if the signal change is greater than a certain threshold, a heartbeat is judged to have occurred, and this process is repeated until the heart rate is calculated. The ECG heart rate graph above illustrates the use of thresholding to calculate the heart rate. This involves identifying different bands (P-wave, R-wave, etc.) and determining

whether a heartbeat has occurred based on the signal change exceeding a certain threshold. This process is repeated until the heart rate is calculated. It should be noted that this method may yield false-positive and false-negative results when confronted with substantial noise or other sources of drift. To circumvent these outcomes, we employ the dynamic thresholding method, which entails utilising the Two-Pass algorithm ($mean + k * standard\ deviation$) to enhance the precision of the heartbeat detection. T. F. Chan and J. G. Lewis, "Computing standard deviations," *Communications of the ACM*, vol. 22, no. 9, pp. 526–531, Sep. 1979, doi: 10.1145/359146.359152.

3. Design

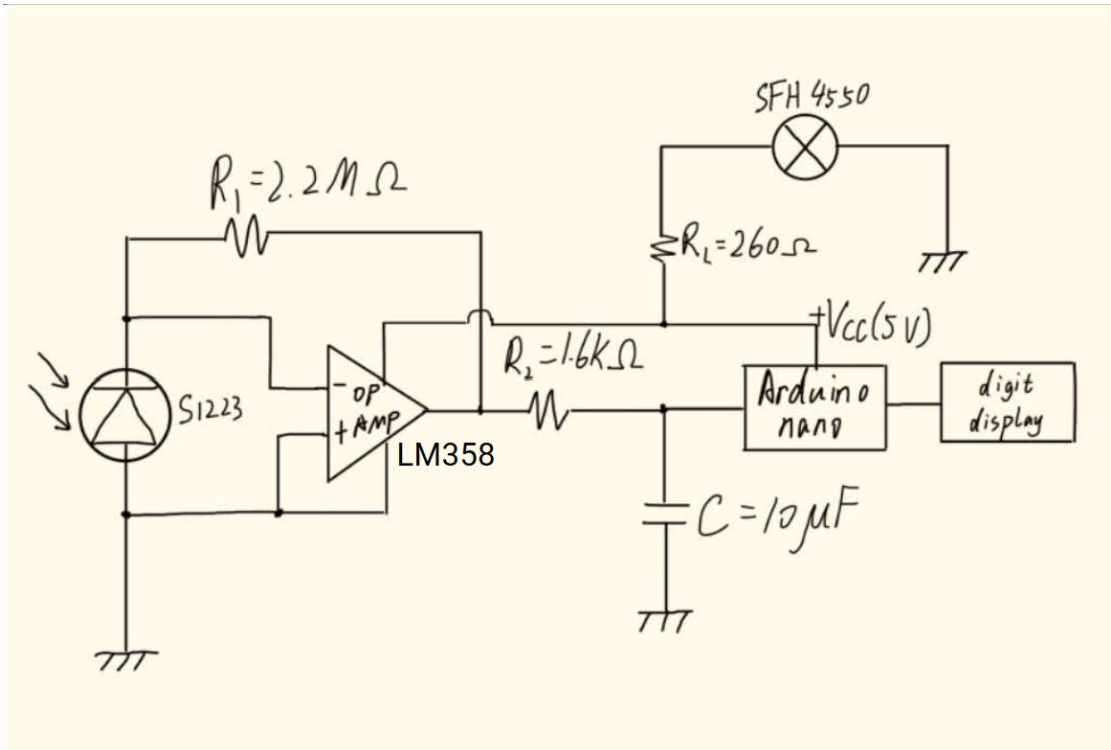
3.1 Specific LED and drive circuit chosen, and why.

The initial consideration when selecting specific LED and driver circuits is the matter of safety. The circuit is designed to contain the diode's optical safety, preventing overcurrent overload. According to the datasheet, it is known to comply with **IEC 62471**. The forward current max is $I_{F\,Max}$ 100mA, indicating that at least $\frac{V_A}{I_F} = 50\ Ohms$ values of resistors should be connected in series. Limiting the current will not damage the LED. "Osram SFH 4550 datasheet - Bing," *Bing*.

https://www.bing.com/search?q=Osram+SFH+4550+datasheet&form=ANNTH1&refig=e6a837a967624096a6ac44bd020c3dfa&pc=LCTS

It is also necessary to consider the concentration of light at a point, which is defined as the half angle of the LED. In order to achieve a high concentration of light with a half angle of 3 degrees, we chose the SFH 4550 LED.Total radiant flux 70Mw.Peak wavelength 860nm.

3.2 Specific photodetector circuit chosen, and why



The circuit in question was selected for the following reasons: a linearly increasing amplified optical voltage with a high amplification gain can be obtained by selecting an appropriate value for R. The

relative simplicity of the circuit prevents signal interference caused by complex circuits and simplifies the troubleshooting process. The modular design is evident with the connection of a filter, indicating the ability to integrate and expand. The low pass filter is designed with the requirement of noise reduction and is required to filter high frequency signals, for this purpose a capacitor with a resistance value of 1.6 kOhms and 10uF is selected, frequencies above 100Hz start to be attenuated significantly. The S1223 series of photodiodes was selected due to its peak sensitivity wavelength of 960 nm, which is closely aligned with that of LEDs (860 nm). This enables the S1223 to exhibit a high specific extinction coefficient, which is beneficial in certain applications. Furthermore, the sensitivity area of the S1223 reaches 2.4 × 2.8 mm, allowing for enhanced light absorption and photocurrent generation.

3.3 Specific data acquisition and processing system chosen, and why.

The optimal data processing system, as determined by the laboratory's equipment, is based on the Arduino platform and Nano board. This system is designed to convert between digital and analogue signals, and the converted signals can be observed in real time on the platform, which is connected to a 4-digit 7-segment

display. The pins controlling the segment are defined as D2-D8, while the pins controlling the digital display are defined as D9-D12.

3.4 Specific algorithm chosen for measuring the pulse rate, and why.

With regard to the algorithm, it is observed that relying solely on the threshold to determine the heartbeat may result in significant inaccuracies (e.g., signal noise, baseline drift, etc.). Consequently, we employ dynamic thresholding as a means of identifying the heartbeat. The advantage of this approach is that the dynamic threshold can be adjusted in real time in accordance with the performance of the signal, which can more effectively eliminate the influence of noise, particularly that of baseline drift. The following section presents the specific code segment in question, with a detailed description provided subsequently.

```
signalSamples[sampleIndex] = signal;
sampleIndex = (sampleIndex + 1) % sampleSize;

calculateMeanAndStdDev(signalSamples, sampleSize, signalMean, signalStdDev);
float dynamicThreshold = signalMean + 2*signalStdDev;
      Core algorithm
if ((signal > dynamicThreshold) && (lastSignal <= dynamicThreshold)) {
  if (currentTime - lastBeatTime > 400) { // Debounce heart beat detection
    beatsPerMinute = 60.0 * 1000.0 / (currentTime - lastBeatTime); // Calculate BPM
    beatsToShow = (int)beatsPerMinute; // Update display BPM
```

The core code is divided into two sections: the first concerns the calculation of the dynamic threshold, while the second addresses the heartbeat judgement and calculation logic. The dynamic threshold method has been presented above, without delving into the underlying details. The following section will focus on the logic. The judgement in question is that, when the value of the heartbeat is greater than the dynamic threshold and the next signal value is less than the dynamic threshold, this indicates that a heartbeat has occurred. This logic is reasonable according to the ECG waveform. It is necessary to adjust the sampling time of each value so that it is neither excessively large nor excessively small (in this case, 10 milliseconds has been chosen). In addition, two judgement time intervals must be established that are reasonable (40 milliseconds has been chosen to prevent noise).

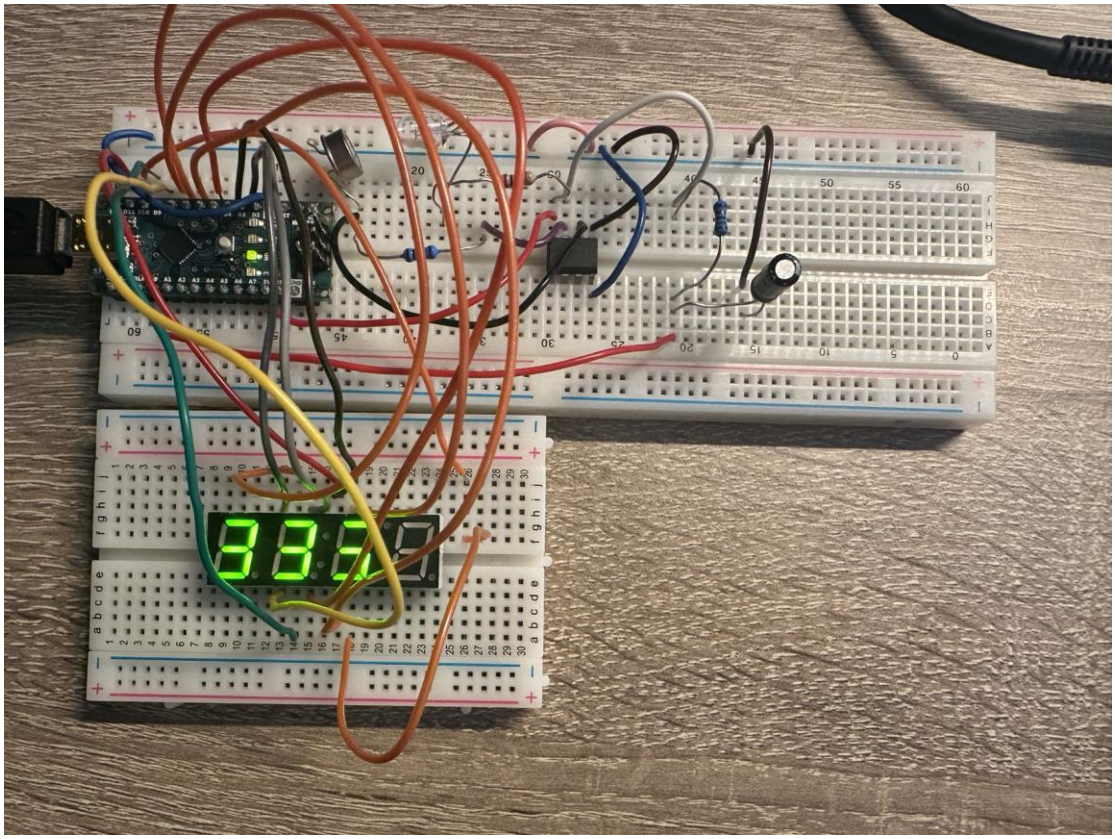
3.5 Demonstration of safety

It is imperative that the light source meets the requisite **IEC 62471** standards and that the circuit is constructed within a reasonable range of current, voltage and power. The drive circuits for LEDs and photodiodes have been designed to comply with the relevant safety standards, as outlined in the preceding section.

4. Construction

4.1 How your circuit is built

In the construction of the circuits, we selected the use of **breadboards and wires**, which offer excellent flexibility for future adjustments. The accompanying images illustrate the process of building the breadboard and the mounting of the LEDs and photodiode.



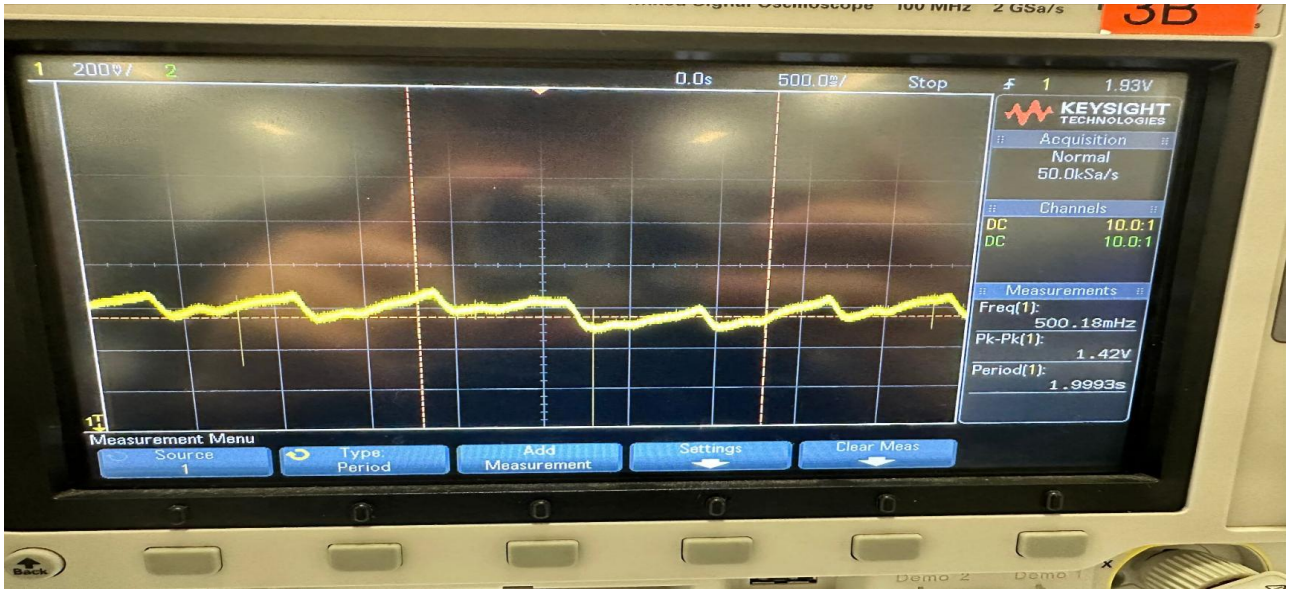
It can be observed that the LED and photodiode are mounted in a manner that allows for the finger to be fixed in place, thereby increasing the amount of light received, which in turn enhances the

accuracy of heartbeat detection. In order to assess the efficacy and convenience of these devices when placed on the human body, we conducted experiments on human tissue (fingers) and fixed the prototype to the human body using coils.

5. Results

5.1 Response to dark conditions

In conditions of low light, the placement of a finger between the LED and the photodiode allows the transmission of a signal to an oscilloscope, which generates a graph of the heartbeat.

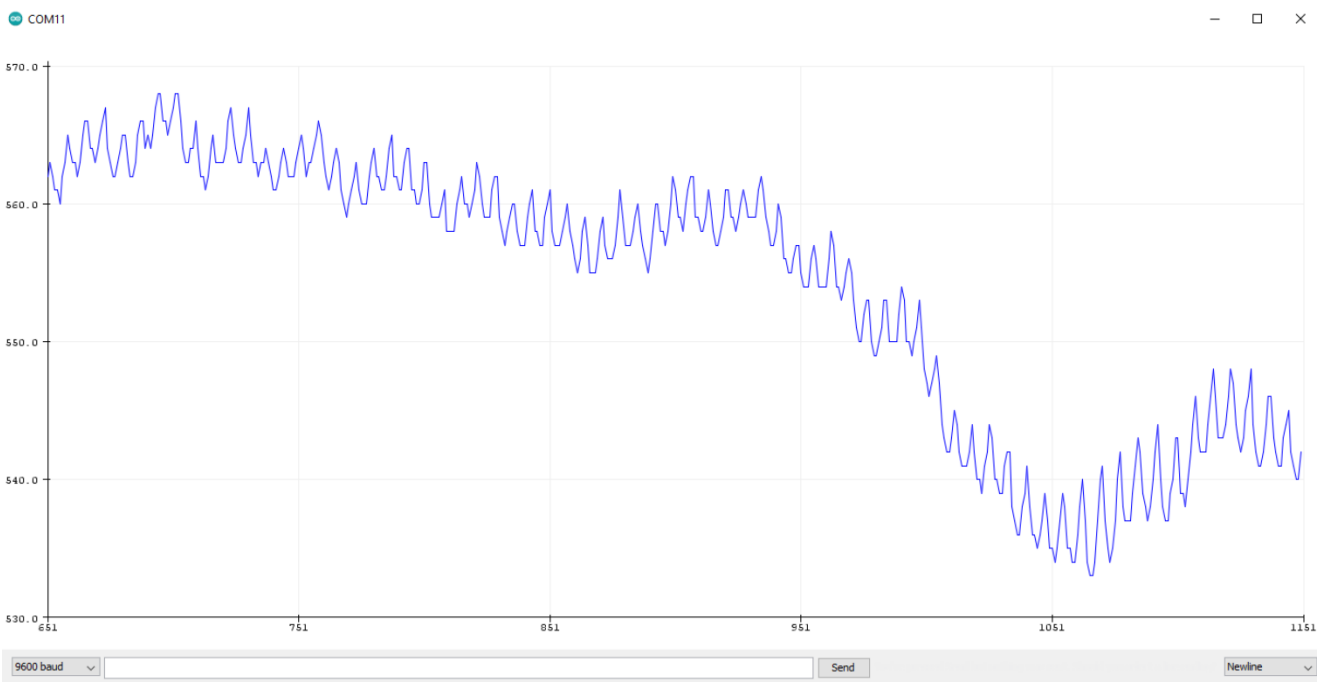


Subsequently, the signals are connected to the previously constructed circuit, and the subject's left hand is connected to the constructed circuit and the right hand is connected to the "Golden Rule" heartbeat detector simultaneously for comparison. Upon

comparison, the self-built circuit exhibited a heart rate of 69 beats per minute, while the Golden Rule heart rate monitor indicated a rate of 62 beats per minute, with a margin of error of 7 beats, or 11 per cent.

5.2 Response to very bright conditions

In the phase of intense ambient light, we observed the generation of the phenomenon of large baseline drift, which has a significant impact on the accuracy of the detection. The following waveform, observed on the Arduino platform, demonstrates a baseline drift of approximately40units.



A second comparison was conducted using the aforementioned methodology. The self-built circuit exhibited a difference of 80 beats,

while the "gold standard" detector exhibited a difference of approximately 67 beats, resulting in a discrepancy of 13 beats. This discrepancy was estimated to be approximately 20 percent, which is less accurate and unable to accurately measure the heart rate.

6. Discussion

It can be stated with certainty that in the dark light environment, our results are not significantly different from those obtained using the standard instrument. This indicates that the successful measurement of the heart rate is possible in the dark light conditions. However, in bright light conditions, the situation is more complex. It can be observed that the presence of baseline drift has a significant impact on our heart rate measurements, resulting in a difference in error of almost 20%. This indicates a failure in the bright light conditions.

The initial and most significant issue is the improper utilisation of filters. Low-pass filters are capable of filtering out high-frequency noise, but are ineffective in eliminating low-frequency noise at a high rate, which ultimately leads to the emergence of baseline drift. The second point is that there is a problem with the algorithm, which requires greater precision in order to determine the time interval and the time interval to remove the noise. In addition, the calculation of

the dynamic threshold requires further attention, particularly with regard to the value of k, which should be adjusted. The third point It is recommended that components generating lower noise, such as LEDs and photodiodes, be selected. Additionally, the instability of the connection due to the use of breadboards can also cause unstable heart rate measurements. A more optimal solution would be to utilise PCB boards or soldered breadboards.

7. Conclusion

The experiments were successful under low-light conditions but unsuccessful under high-light conditions. The project demonstrated that biology and electronics can be combined in a close manner, and the use of electrical signals to detect biological signals represents a promising area for future development. The project consolidated the noise of analogue electronics, the use of operational amplifiers and the use of oscilloscopes. The utilisation of coding for embedded systems permitted the calculation of heartbeats, while a more profound comprehension of the utilisation of the Arduino was attained. Additionally, a more profound comprehension of the concepts inherent to the discipline of engineering was achieved, with a constant striving to achieve a balance between theoretical and practical considerations. Should time permit, further investigation will be conducted into the effects

of filters and photoreceptors on the results, with a view to optimising the algorithms and developing solutions that combine convenience with accuracy.

Reference

1. Wikipedia contributors, “Electrocardiography,” *Wikipedia*, Apr. 09, 2024. https://en.wikipedia.org/wiki/Electrocardiography

2. T. F. Chan and J. G. Lewis, “Computing standard deviations,” *Communications of the ACM*, vol. 22, no. 9, pp. 526–531, Sep. 1979, doi: 10.1145/359146.359152.

3. “Osram SFH 4550 datasheet - Bing,” *Bing*. https://www.bing.com/search?q=Osram+SFH+4550+datasheet&form=ANNTH1&refig=e6a837a967624096a6ac44bd020c3dfa&pc=LCTS